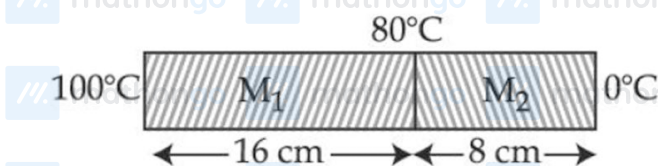


Q1 - 24 June - Shift 1

Two metallic blocks M_1 and M_2 of same area of cross-section are connected to each other (as shown in figure). If the thermal conductivity of M_2 is K then the thermal conductivity of M_1 will be : [Assume steady state heat conduction]



- (A) 10 K (B) 8 K
(C) 12.5 K (D) 2 K

Space for your notes:

Q2 - 24 June - Shift 2

A 100 g of iron nail is hit by a 1.5 kg hammer striking at a velocity of 60 ms^{-1} . What will be the rise in the temperature of the nail if one fourth of energy of the hammer goes into heating the nail? [Specific heat capacity of iron = $0.42 \text{ Jg}^{-1} \text{ }^\circ\text{C}^{-1}$]

- (A) 675°C (B) 1600°C
(C) 160.7°C (D) 6.75°C

Space for your notes:

Q3 - 24 June - Shift 2

An electric bulb is rated as 200 W. What will be the peak magnetic field at 4 m distance produced by the radiations coming from this bulb? Consider this bulb as a point source with 3.5% efficiency.

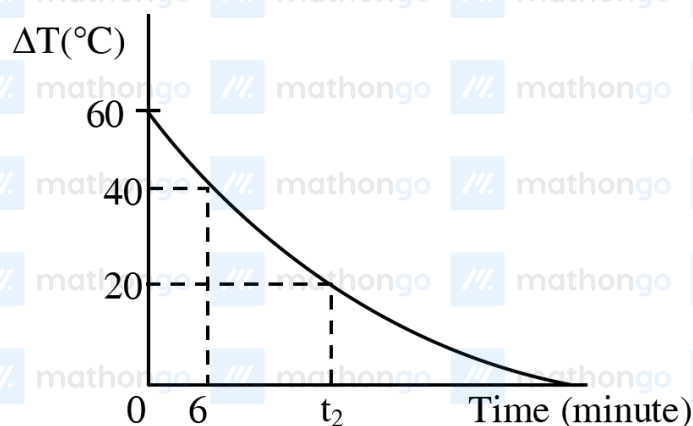
- (A) $1.19 \times 10^{-8} \text{ T}$ (B) $1.71 \times 10^{-8} \text{ T}$
(C) $0.84 \times 10^{-8} \text{ T}$ (D) $3.36 \times 10^{-8} \text{ T}$

Space for your notes:

Q4 - 24 June - Shift 2

In an experiment to verify Newton's law of cooling, a graph is plotted between the temperature difference (ΔT) of the water and surroundings and time as shown in figure. The initial temperature of water is taken as 80°C . The value of t_2 as mentioned in the graph will be _____.

Space for your notes:

**Q5 - 25 June - Shift 1**

A steam engine intakes 50g of steam at 100°C per minute and cools it down to 20°C . If latent heat of vaporization of steam is 540 cal g^{-1} , then the heat rejected by the steam engine per minute is _____ $\times 10^3 \text{ cal}$.

Space for your notes:

Q6 - 25 June - Shift 2

Questions

MathonGo

A solid metallic cube having total surface area 24 m^2 is uniformly heated. If its temperature is increased by 10°C , calculate the increase in volume of the cube (Given: $\alpha = 5.0 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$)

(A) $2.4 \times 10^6 \text{ cm}^3$

(B) $1.2 \times 10^5 \text{ cm}^3$

(C) $6.0 \times 10^4 \text{ cm}^3$

(D) $4.8 \times 10^5 \text{ cm}^3$

Q7 - 25 June - Shift 2

A copper block of mass 5.0 kg is heated to a temperature of 500°C and is placed on a large ice block. What is the maximum amount of ice that can melt? [Specific heat of copper: $0.39 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$ and latent heat of fusion of water : 335 J g^{-1}]

(A) 1.5 kg

(B) 5.8 kg

(C) 2.9 kg

(D) 3.8 kg

Q8 - 26 June - Shift 2

A geyser heats water flowing at a rate of $2.0 \text{ kg per minute}$ from 30°C to 70°C . If geyser operates on a gas burner, the rate of combustion of fuel will be _____ g min^{-1}

[Heat of combustion = $8 \times 10^3 \text{ J g}^{-1}$

Specific heat of water = $4.2 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$]

#MathBoleTohMathonGo

Q9 - 27 June - Shift 2

A lead bullet penetrates into a solid object and melts. Assuming that 40% of its kinetic energy is used to heat it, the initial speed of bullet is:

(Given, initial temperature of the bullet = 127°C ,

Melting point of the bullet = 327°C ,

Latent heat of fusion of lead = $2.5 \times 10^4 \text{ J Kg}^{-1}$,

Specific heat capacity of lead = 125 J/kg K)

(A) 125 ms^{-1}

(B) 500 ms^{-1}

(C) 250 ms^{-1}

(D) 600 ms^{-1}

Space for your notes:

Q10 - 28 June - Shift 2

Resistance of the wire is measured as 2Ω and 3Ω at 10°C and 30°C respectively. Temperature co-efficient of resistance of the material of the wire is :

(A) $0.033^{\circ}\text{C}^{-1}$

(B) $-0.033^{\circ}\text{C}^{-1}$

(C) $0.011^{\circ}\text{C}^{-1}$

(D) $0.055^{\circ}\text{C}^{-1}$

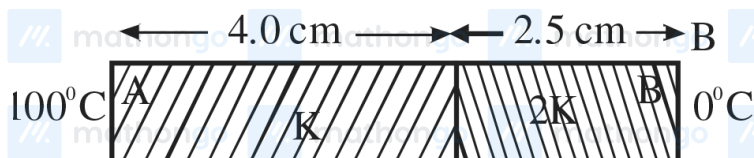
Space for your notes:

Q11 - 29 June - Shift 1

Questions

MathonGo

As per the given figure, two plates A and B of thermal conductivity K and $2K$ are joined together to form a compound plate. The thickness of plates are 4.0 cm and 2.5 cm respectively and the area of cross-section is 120 cm^2 for each plate. The equivalent thermal conductivity of the compound plate is $\left(1 + \frac{5}{\alpha}\right)K$, then the value of α will be _____.



Space for your notes:

Q12 - 29 June - Shift 2

At what temperature a gold ring of diameter 6.230 cm be heated so that it can be fitted on a wooden bangle of diameter 6.241 cm ? Both the diameters have been measured at room temperature (27°C).

(Given: coefficient of linear thermal expansion of gold $\alpha_L = 1.4 \times 10^{-5}\text{ K}^{-1}$)

(A) 125.7°C (B) 91.7°C (C) 425.7° (D) 152.7°C

Space for your notes:

Answer Key

Q1 (B)

Q2 (C)

Q3 (B)

Q4 (16)

Q5 (31)

Q6 (B)

Q7 (C)

Q8 (42)

Q9 (B)

Q10 (A)

Q11 (21)

Q12 (D)

#MathBoleTohMathonGo

Q1 (B)

$$\Delta T \propto R \propto \frac{\ell}{k},$$

$$\frac{\Delta T_1}{\Delta T_2} = \frac{\ell_1}{k_1} \times \frac{k_2}{\ell_2} = \frac{16}{k_1} \times \frac{k}{8}$$

$$\frac{20}{80} = \frac{16}{k_1} \times \frac{k}{8} \rightarrow k_1 = 8k$$

Q2 (C)

$$\frac{1}{2} \times 1.5 \times 60^2 \times \frac{1}{4} = 0.1 \times 420 \times \Delta T$$

Q3 (B)

$$\frac{\eta P}{4\pi r^2} = \frac{cB_0^2}{2\mu_0}$$

$$B_0 = \sqrt{\frac{\mu_0}{4\pi} \frac{\eta P}{c} \frac{1}{r}}$$

$$\Rightarrow B_0 = \frac{1}{4} \sqrt{\frac{10^{-7} \times 4 \times 3.5}{3 \times 10^8}} = 1.71 \times 10^{-8} \text{ T}$$

Q4 (16)

$$T - T_0 = (T_i - T_0) e^{-\frac{Bt}{ms}}$$

$$6\lambda = \ln 1.5$$

$$40 = 60e^{-\lambda(6)} \Rightarrow 6\lambda = \ln 1.5$$

$$20 = 60e^{-\lambda t_2} \Rightarrow t_2 \lambda = \ln 3$$

$$\frac{t_2}{6} = \frac{\ln 3}{\ln 1.5}$$

$$\therefore t_2 = 16.25 \text{ min}$$

$$\text{So } \approx 16$$

Q5 (31)

$$\text{Heat rejected} = mL_f + mS\Delta T$$

$$= (50 \times 540) + 50 (1) (100 - 20)$$

$$= 31000 \text{ Cal}$$

$$= 31 \times 10^3 \text{ Cal}$$

Q6 (B)

$$\text{Increase in volume } \Delta V = \gamma V_0 \Delta T$$

$$\gamma = 3\alpha$$

$$\text{So } \Delta V = (3\alpha) V_0 \Delta T$$

$$\text{Total surface area} = 6a^2, \text{ where } a \text{ is side length}$$

$$24 = 6a^2 \quad a = 2\text{m}$$

$$\text{Volume } V_0 = (2)^3 = 8\text{m}^3$$

$$\Delta V = (3 \times 5 \times 10^{-4})(8) \times 10$$

$$= 1.2 \times 10^5 \text{cm}^3$$

Q7 (C)Heat given by block to get 0°C temperature

$$\Delta Q_1 = 5 \times (0.39 \times 10^3) \times (500 - 0)$$

$$= 975 \times 10^3 \text{J}$$

Heat absorbed by ice to melt m mass

$$\Delta Q_2 = m \times (335 \times 10^3) \text{J}$$

$$\Delta Q_1 = \Delta Q_2$$

$$m \times (335 \times 10^3) = 975 \times 10^3$$

$$m = \frac{975}{335} = 2.910 \text{kg}$$

Q8 (42)

$$m = 2000 \text{ gm/min}$$

Heat required by water/min = $m\Delta T$

$$= (2000) \times 4.2 \times 40 \text{ J/min}$$

$$= 336000 \text{ J/min}$$

$$\text{The rate of combustion} = \left(\frac{dm}{dt} L \right) = 336000 \text{ J/min}$$

$$\frac{dm}{dt} = \frac{336000}{8 \times 10^3} \text{ g/min}$$

$$= 42 \text{ gm/min}$$

#MathBoleTohMathonGo

Q9 (B)

$$m \times 125 \times 200 + m \times 2.5 \times 10^4 = \frac{1}{2}mv^2 \times \frac{40}{100}$$

$$V = 500 \text{ m/s}$$

Q10 (A)

$$R = R_0(1 + \alpha\Delta T)$$

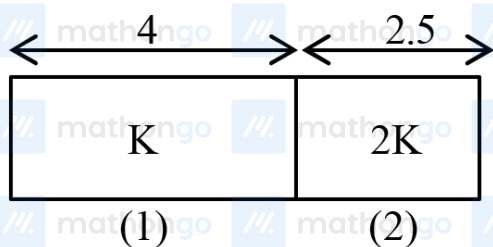
$$3 = R_0(1 + \alpha(30 - 0))$$

$$2 = R_0(1 + \alpha(10 - 0))$$

$$\frac{3}{2} = \frac{1 + 30\alpha}{1 + 10\alpha}$$

$$\alpha = \frac{1}{30} = 0.033$$

Q11 (21)



$$\frac{\Delta Q}{\Delta t} = \left(\frac{1}{R} \right) \Delta T$$

R : Thermal resistivity

$$\therefore R_1 = \frac{L_1}{K_1 A} = \frac{L_1}{K(120)}$$

$$L_1 = 4 \text{ cm}$$

$$A = 120 \text{ cm}^2$$

$$R_2 = \frac{2.5}{(2K)(120)}$$

Now, R_{eq} of this series combination

$$R_{eq} = R_1 + R_2$$

$$\text{where } L_{eq} = 4 + 2.5 = 6.5$$

$$\frac{L_{eq}}{K_{eq}(A)} = \frac{4}{K(120)} + \frac{5}{2K(120)}$$

$$\frac{6.5}{K_{eq}(120)} = \frac{4}{K(120)} + \frac{5}{4K(120)}$$

$$\frac{6.5}{K_{eq}} = \frac{21}{4K}$$

$$K_{eq} = \frac{26}{21} K = \left(1 + \frac{5}{21} \right) K$$

$$\therefore a = 21$$

Q12 (D)

$$\Delta \ell = 6.241 - 6.230 = 0.011 \text{ cm}$$

$$\Delta \ell = \ell \alpha \Delta \theta$$

$$0.011 = 6.230 \times 1.4 \times 10^{-5} (\theta - 27)$$

$$\theta - 27 = \frac{0.011 \times 10^5}{6.230 \times 1.4}$$

$$\theta \approx 153.11 \text{ nearest is } 152.7^\circ\text{C}.$$